

Mohammadmahdi (Mahdy) Maharebi

M.Sc. Student – Electrical Communication Engineering

Kassel, Germany

contact@mahdymahareb.de | +49 178 1780777

GitHub: github.com/mmaharebi | Website: mahdymahareb.de | LinkedIn: linkedin.com/in/mmaharebi

Professional Summary

M.Sc. student with a **solid analytical and computational background** in electromagnetics, RF/microwave engineering, computational photonics, scientific computing, and signal processing. Experienced in developing and validating numerical solvers (FDTD, FEM, eigenmode methods) and building documented simulation software with reproducible research workflows. Silver medalist in Iran's National Physics Olympiad. Interested in research-oriented roles, HiWi positions, scientific software development, and Master's thesis opportunities in electromagnetics, simulation, antennas, or optical modelling.

Education

M.Sc. Electrical Communication Engineering

2024–Present

University of Kassel, Germany

GPA: 1.53/1.0 (sehr gut)

Key courses: Microwaves (1.0), Optoelectronics (1.0), Engineering Mathematics (1.0), Digital Communications Lab (1.3), Digital Communications (1.7).

Focus areas: computational EM, photonics, optimization, signal processing.

B.Sc. Electrical Engineering

2017–2023

Sharif University of Technology, Iran

Focus on electromagnetics, microwave engineering, and signal processing.

Teaching Assistant: Circuit Analysis I (Head TA), Circuit Analysis II, Object-Oriented Programming in Java.

Research Experience

2D FDTD PEC Cavity Solver

2025

Python/NumPy – Computational Electromagnetics

- Developed TMz FDTD solver with CFL-stable update equations.
- Validated five cavity eigenmodes with 0.43% mean error.
- Implemented FFT-based mode extraction with probe averaging.
- Produced convergence plots and animations in a reproducible workflow.

Fiber Mode Perturbation Analysis

2025

Python/SciPy – Operator Theory & Numerical Methods

- Derived first-order sensitivities of β and n_{eff} under fabrication variations.
- Implemented eigenmode solvers with $< 0.1\%$ numerical error.
- Produced tolerance maps for realistic refractive-index and radius deviations.
- Developed manuscript-ready figures and reproducible analysis scripts.

Selected Engineering Projects

Symulation – Interactive C++ Simulation Lab

2026–Present

C++17 / CMake / raylib / ImGui / ImPlot / ImPlot3D / Eigen / CGAL

- Built a native desktop simulation lab for physics, electromagnetics, and numerical visualization with real-time GUI controls.
- Designed a modular simulation architecture covering BCC spring-mass wave propagation, Hertzian dipole fields, thin-wire dipole MoM/Pocklington demo, double pendulum, three-body gravity, and Helmholtz FEM eigenmodes on editable Bézier domains.
- Prepared the project for public release with Git submodules, clean build instructions, GIF demos, technical notes, MIT license, GitHub documentation, portfolio integration, and a technical blog post.
- Code: github.com/mmaharebi/symulation; Blog: mahdymahareb.de/blog/symulation.

Antenna & RF Design Portfolio

2021–2023

HFSS / ADS / MATLAB

- Designed a 16 dBi horn antenna with optimized matching.
- Modeled a phased dipole array with mutual coupling analysis.
- Designed a 3 dB branch-line coupler with >20 dB isolation.

Adaptive FEM Visualization Tool

2023

C# / WPF

- Implemented a tool for FEM field visualization and mesh reconstruction.
- Integrated Delaunay triangulation and automated parsing routines.

Technical Skills

Computational EM: FDTD, FEM post-processing, eigenmode solvers, perturbation analysis

Scientific Software: C++17, CMake, Python, Eigen, CGAL, ImGui, ImPlot

RF Tools: HFSS, ADS; S-parameters, array factor analysis

Programming: Python, MATLAB, C++, C#, TypeScript/React

Research Tools: LaTeX, Git/GitHub, Jupyter, Linux/Bash

Domains: EM simulation, photonics, antennas, microwaves, signal processing, 5G/6G

Awards & Distinctions

Silver Medal – National Physics Olympiad

2017

Iran

National Elites Foundation Recognition

2017

Government of Iran

Languages

English: Professional

German: A2 (actively improving)

Persian: Native